Alice's father sighed. "Fine. I'm not saying you're convincing me of anything, but I don't want to get bogged down in philosophical hair-splitting. Now that you mention it, Popper's question does seem kind of reasonable. Why don't worlds fuse together as well as branch apart? If we have a spin that is an equal superposition of up and down, we can predict the probability of observing either outcome if we do a measurement in the future. But if we have a spin that is purely up, and we are told that it was just measured, we have absolutely no way of knowing what kind of superposition it was in pre-measurement (except that it wasn't purely down). Where does the difference come from?"

Alice seemed ready for this one. "That's just thermodynamics, really. Or at least, it's the arrow of time, pointing from the past to the future. We remember yesterday but not tomorrow; cream and coffee mix together but they don't spontaneously unmix. Wave functions branch, but don't unbranch."

"Sounds suspiciously circular. As I understand it, one of the purported features of Many-Worlds is that wave functions only obey the Schrödinger equation; there's no separate collapse postulate. Back when I learned quantum mechanics, we knew that wave functions collapsed toward the future and not toward the past, and that was part of the assumptions. I don't see why that should still be true for Everett, where the Schrödinger equation is completely reversible. What do cream and coffee have to do with wave functions?"

Alice nodded. "Perfectly good question. Let's set the stage a bit. The second law of thermodynamics posits that entropy—roughly, the disorderliness or randomness of a configuration, as you know—never decreases in closed systems. Ludwig Boltzmann explained this back in the 1870s. Entropy counts the number of ways that atoms can be arranged so that the system looks the same from a macroscopic perspective. The reason why it increases is simply that there are many more ways to be high-entropy than to be low-entropy, so it's improbable that entropy would ever go down. Right?"

"Sure," her father agreed. "But that's all classical; Boltzmann didn't know anything about quantum mechanics."

"Right, but the basic idea is the same. Boltzmann explained why entropy tends to increase, but he didn't give a reason why it was ever low in the first place. These days we appreciate that it is a cosmological fact that